

Real-Time Clock



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The clock described here displays the current time and date, and can be built within a small budget using AT89C52 microcontroller. It also has the facility to set an alarm and is visible even in the dark. This construction project will help you understand concepts like real-time use of microcontrollers and scanning related to displays. The device is dual powered and switches to battery supply when the mains power is 'off.'

Circuit description

Fig. 1 shows the circuit of the real-time clock and Fig. 3 shows the power supply section. The clock is built around microcontroller AT89C52 (IC1), LTS542 seven-segment displays (DIS1 through DIS6), regulator 7805 (IC2) and a few discrete components.

PARTS LIST

Semiconductors:

- IC1 - AT89C52 microcontroller
- IC2 - 7805, 5V regulator
- T1-T8 - BC557 npn transistor
- D1-D4 - 1N4007 rectifier diode
- DIS1-DIS6 - LTS542 common-anode 7-segment display
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, 45% carbon):

- R1-R5 - 10-kilo-ohm
- R6-R13 - 100-ohm
- R14-R21 - 1-kilo-ohm
- R22 - 470-ohm
- RNW1 - 10-kilo-ohm resistor network

Capacitors:

- C1 - 10µF, 16V electrolytic
- C2, C3 - 33pF ceramic disk
- C4 - 1000µF, 35V electrolytic
- C5 - 0.1µF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 12V, 500mA secondary transformer
- X_{1AL} - 12MHz crystal
- S1-S5 - Push-to-on tactile switch
- PZ1 - Piezobuzzer

The AT89C52 is a low-power, high-performance CMOS 8-bit microcontroller. It has 8 kB of Flash, 256 bytes of RAM, 32 input/output (I/O) lines, three 16-bit timers/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry. Out of its four I/O ports, ports P0 and P2 are used to drive seven-segment displays (DIS1 through DIS6), LEDs (LED1 and LED2) and piezobuzzer PZ1.

Port pins P2.0 through P2.7 of the microcontroller are connected to seven-segment displays as shown in Fig. 1. To make the PCB layout easier, DIS1, DIS3 and DIS5 are connected differently from DIS2, DIS4 and DIS6. Pins e, d, c, dp, b, a, f and g of displays DIS1, DIS3 and DIS5 are connected to port pins P2.0 through P2.7, respectively. In case of DIS2, DIS4 and DIS6, pins dp, c, d, e, g,

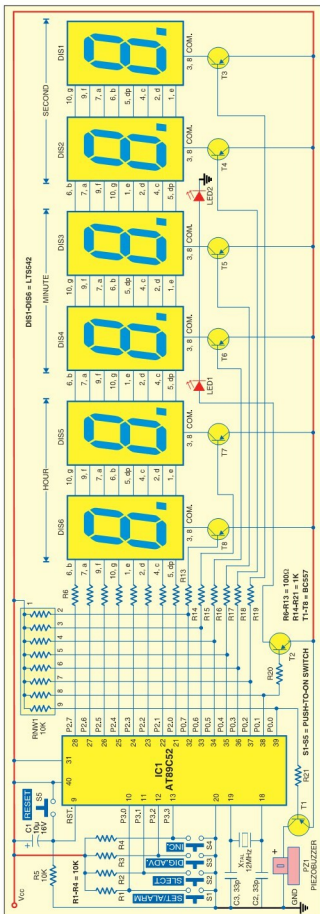


Fig. 1: Circuit of the real-time clock